CADT

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Institute of Digital Technology

Database Project Proposal

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I. Project Topic Movie streaming online

NaekWatch is a movie streaming online which provides user movie and TV-show.

II. ENTITY and Attribute

Account: Represents the registered users of the platform.

- Attributes: Email (Unique ID), Username, Password Hash (Security), Full Name, Phone Number, Date of Birth, Total Points (Gamification balance), and Account Status (Active/Banned).

Redemption: Tracks the exchange of points for physical or digital rewards.

- Attributes**:** Points Used, Redemption Date, and Status (Pending/Approved).

Content: The core library consisting of movies and series.

- Attributes**:** Title, Type (Movie/Series), Director, Release Year, Rating (0–10 scale), Duration, and Price (for premium rentals).

Genre: Categorization labels for discovery.

- Attributes**:** genre_ID, genre_name (e.g., Action, Sci-Fi, Trending).

Actor: Profiles of individuals appearing in the content.

- Attributes**:** actor_ID, actor_name.

Watch History: Logs every piece of content a user has viewed.

- Attributes**:** watch_ID, account_ID, content_ID, watched_at.

Watchlist: A personalized "Save for Later" list for users.

- Attributes**:** Watchlist_ID, account_ID, content_ID, added at.

Movie Purchase: Financial records of content bought by users.

- Attributes**:** purchase_ID(PK), account_ID(FK) purchase_at, content_ID, transaction_ID, amount_paid, payment_status.

Task: The master list of challenges available to users.

- Attributes**:** task_ID, task_name, description, required_movie_count, reward_point, create_time.

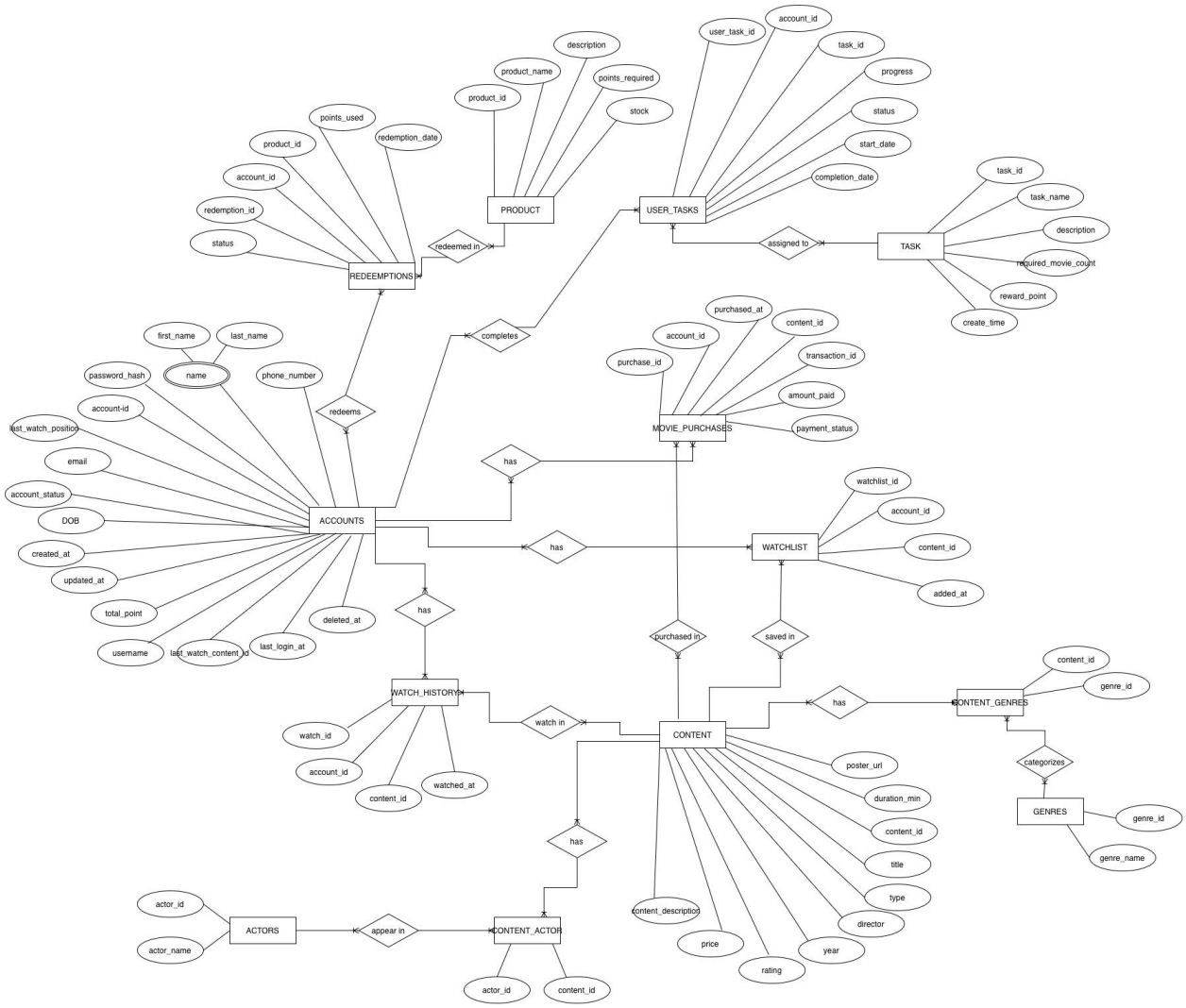
User task: The bridge tracking an individual's progress on a specific challenge.

- Attributes**:** user_task_ID(PK), account_ID, task_ID, progress, task_status, start_date, complete_date.

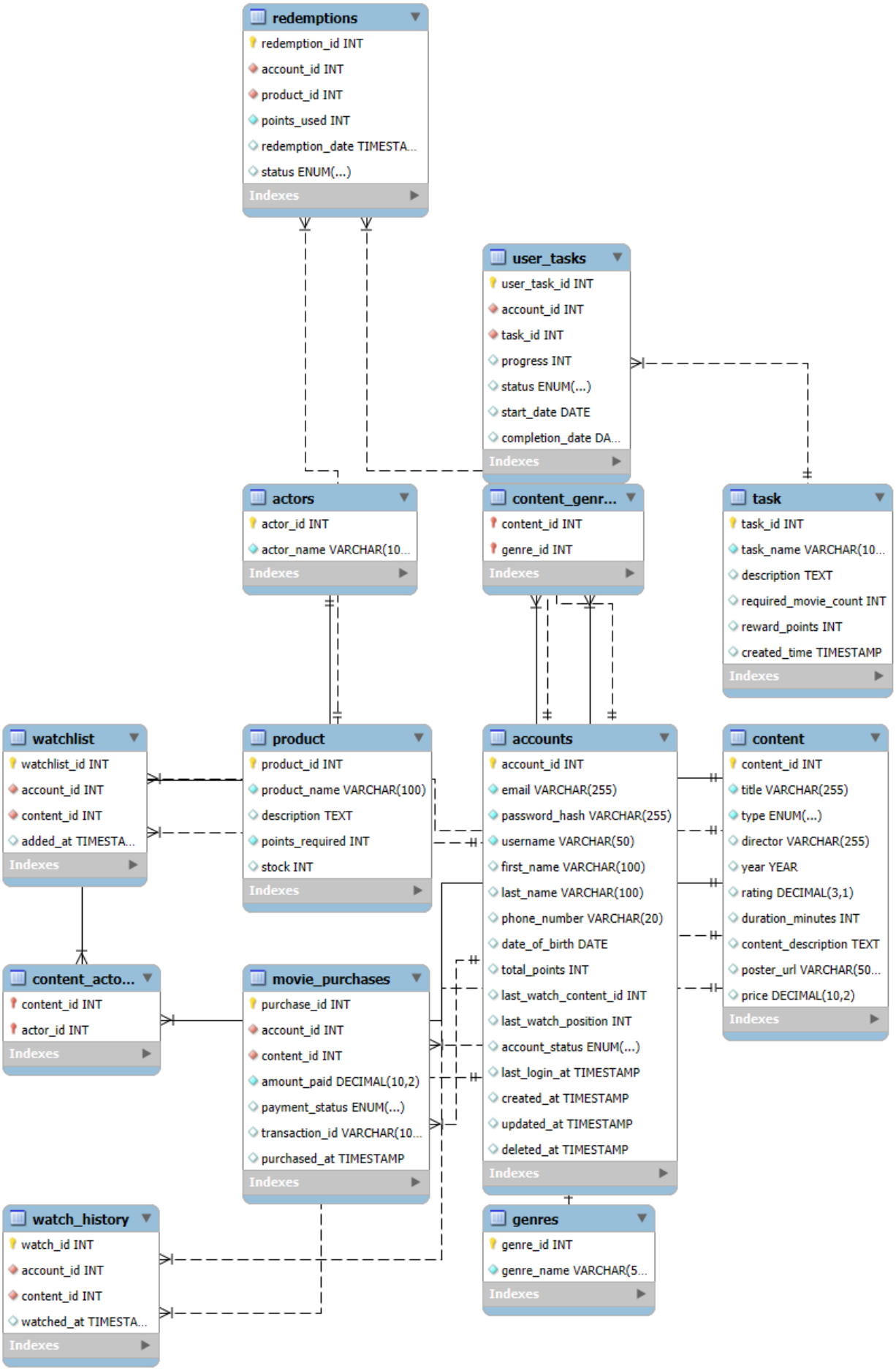
Product: Physical merchandise available in the reward shop.

- Attributes**:** product_ID, product_name, description, points_required, stock.

III. ERD



IV. Relation Model


redemptions
redemption_id INT
account_id INT
product_id INT
points_used INT
redemption_date TIMESTA...
status ENUM(...)
Indexes
user_tasks
user_task_id INT
account_id INT
task_id INT
progress INT
status ENUM(...)
start_date DATE
completion_date DA...
Indexes
actors
actor_id INT
actor_name VARCHAR(10...
Indexes
content_genr...
content_id INT
genre_id INT
Indexes
task
task_id INT
task_name VARCHAR(10...
description TEXT
required_movie_count INT
reward_points INT
created_time TIMESTAMP
Indexes
watchlist
watchlist_id INT
account_id INT
content_id INT
added_at TIMESTA...
Indexes
product
product_id INT
product_name VARCHAR(100)
description TEXT
points_required INT
stock INT
Indexes
accounts
account_id INT
email VARCHAR(255)
password_hash VARCHAR(255)
username VARCHAR(50)
first_name VARCHAR(100)
last_name VARCHAR(100)
phone_number VARCHAR(20)
date_of_birth DATE
total_points INT
last_watch_content_id INT
last_watch_position INT
account_status ENUM(...)
last_login_at TIMESTAMP
created_at TIMESTAMP
updated_at TIMESTAMP
deleted_at TIMESTAMP
Indexes
content
content_id INT
title VARCHAR(255)
type ENUM(...)
director VARCHAR(255)
year YEAR
rating DECIMAL(3,1)
duration_minutes INT
content_description TEXT
poster_url VARCHAR(50...
price DECIMAL(10,2)
Indexes
content_acto...
content_id INT
actor_id INT
Indexes
movie_purchases
purchase_id INT
account_id INT
content_id INT
amount_paid DECIMAL(10,2)
payment_status ENUM(...)
transaction_id VARCHAR(10...
purchased_at TIMESTAMP
Indexes
watch_history
watch_id INT
account_id INT
content_id INT
watched_at TIMESTA...
Indexes
genres
genre_id INT
genre_name VARCHAR(5...
Indexes